

## 2020 AMC 10B Problem 8 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

### 2020 AMC 10B Problem 8

**Problem:**

Points  $P$  and  $Q$  lie in a plane with  $PQ = 8$ . How many locations for point  $R$  in this plane are there such that the triangle with vertices  $P$ ,  $Q$ , and  $R$  is a right triangle with area 12 square units?

**Answer Choices:**

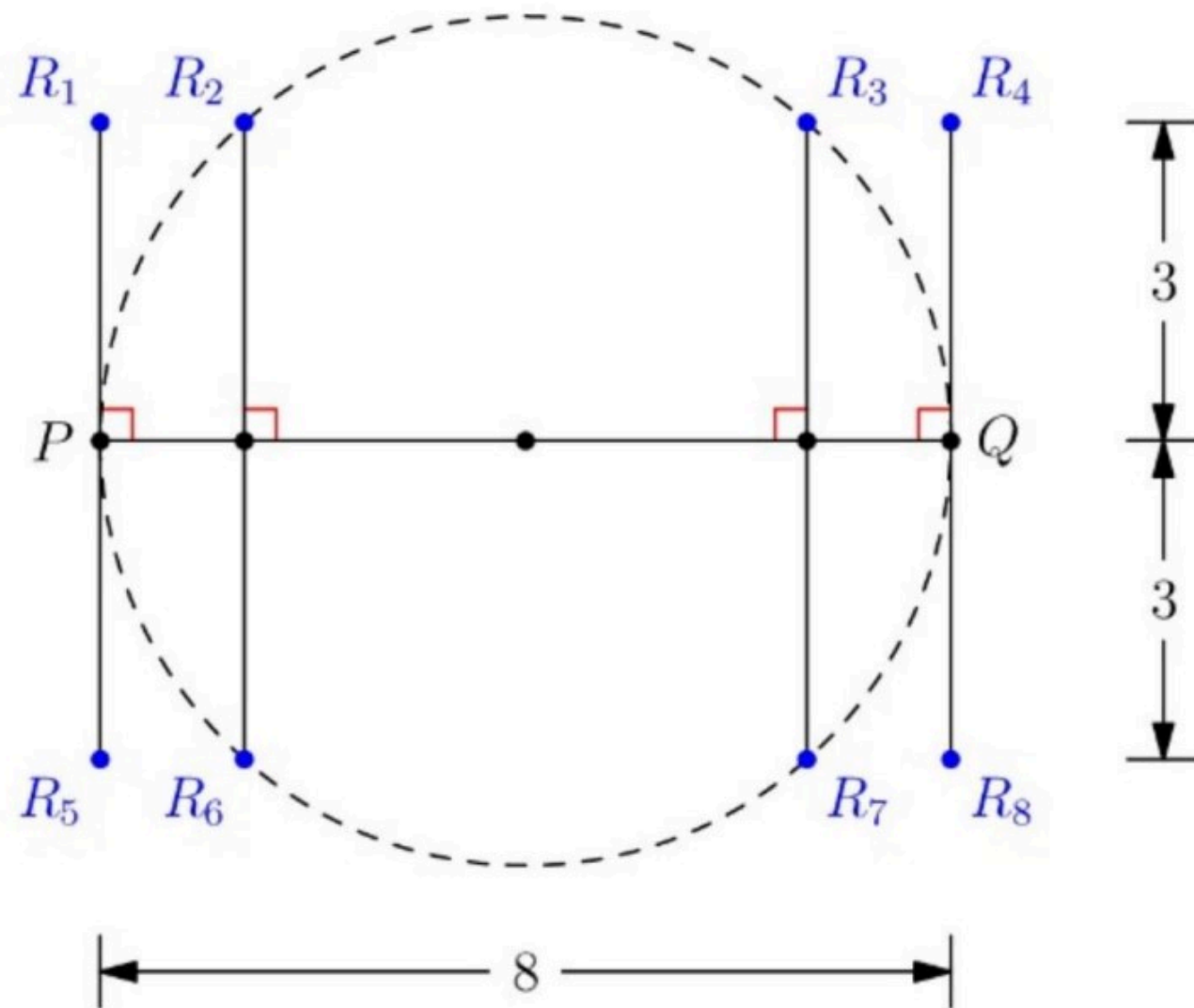
- A. 2
- B. 4
- C. 6
- D. 8
- E. 12

**Solution:**

Let the brackets denote areas. We are given that

$$[PQR] = \frac{1}{2} \cdot PQ \cdot h_R = 12$$

Since  $PQ = 8$ , it follows that  $h_R = 3$ . We construct a circle with diameter  $\overline{PQ}$ . All such locations for  $R$  are shown below:



We apply casework to the right angle of  $\triangle PQR$ :

1. If  $\angle P = 90^\circ$ , then  $R \in \{R_1, R_5\}$  by the tangent.
2. If  $\angle Q = 90^\circ$ , then  $R \in \{R_4, R_8\}$  by the tangent.
3. If  $\angle R = 90^\circ$ , then  $R \in \{R_2, R_3, R_6, R_7\}$  by the Inscribed Angle Theorem.

Together, there are  $\boxed{(D) 8}$  such locations for  $R$ .

**Remarks:**

The reflections of  $R_1, R_2, R_3, R_4$  about  $\overleftrightarrow{PQ}$  are  $R_5, R_6, R_7, R_8$ , respectively

The reflections of  $R_1, R_2, R_5, R_6$  about the perpendicular bisector of  $\overline{PQ}$  are  $R_4, R_3, R_8, R_7$ , respectively.

**OR**

Let the brackets denote areas. We are given that

$$[PQR] = \frac{1}{2} \cdot PQ \cdot h_R = 12$$

Since  $PQ = 8$ , it follows that  $h_R = 3$ . Without the loss of generality, let  $P = (-4, 0)$  and  $Q = (4, 0)$ . We conclude that the  $y$ -coordinate of  $R$  must be  $\pm 3$ .

We apply casework to the right angle of  $\triangle PQR$ :

1.  $\angle P = 90^\circ$ . The  $x$ -coordinate of  $R$  must be  $-4$ , so we have  $R = (-4, \pm 3)$ . In this case, there are 2 such locations for  $R$ .
2.  $\angle Q = 90^\circ$ . The  $x$ -coordinate of  $R$  must be  $4$ , so we have  $R = (4, \pm 3)$ . In this case, there are 2 such locations for  $R$ .
3.  $\angle R = 90^\circ$ . For  $R = (x, 3)$ , the Pythagorean Theorem  $PR^2 + QR^2 = PQ^2$  gives

$$[(x+4)^2 + 3^2] + [(x-4)^2 + 3^2] = 8^2$$

Solving this equation, we have  $x = \pm\sqrt{7}$ , or  $R = (\pm\sqrt{7}, 3)$ . For  $R = (x, -3)$ , we have  $R = (\pm\sqrt{7}, -3)$  by a similar process. In this case, there are 4 such locations for  $R$ .

Together, there are  $2 + 2 + 4 = \boxed{(D) 8}$  such locations for  $R$ .

*The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) ↗*

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